

Performance Testing

Date	29 September 2026 – 2 October 2026
Team ID	6ab4fddea238dc999eb665e9
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how PocketSmart AI behaves under expected and peak load conditions. The testing evidence covers the FastAPI application, planner APIs, recommendation history and AI-assisted recommendation flows.

Step 1: Testing Overview

Field	Details
Tool Used	Postman and Apache JMeter
Testing	Load Testing and basic Stress Testing
Module / API	User Registration, User Login, Home Interior Recommendations, Party Planner Recommendations, Jewelry Recommendation History
Environment	Local FastAPI / Uvicorn development environment
Duration	29 September 2026 – 2 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Concurrent user registration and login requests.	5	60	Successful authentication responses without unexpected errors.
2	Concurrent Home Interior recommendation requests using valid budget and preferences.	5	60	Planner API remains responsive and returns valid recommendations.
3	Concurrent Party Planner and Jewelry Planner recommendation requests.	5	60	Planner APIs remain responsive with valid recommendation responses.
4	Repeated Recommendation History retrieval after recommendation generation.	10	60	Stable history retrieval with no database or request failures.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	842 ms	Pass	Postman collection average response time.
2	Response Time (Max)	< 5 seconds	2.56 sec	Pass	Maximum shown in JMeter summary.
3	Throughput (Req/sec)	To be measured	13.1 req/sec	Pass	JMeter total throughput shown in Summary Report.
4	Error Rate	< 1%	0.08%	Pass	JMeter total error rate shown in Summary Report.
5	CPU Utilization	< 80%	Not shown	Pending	CPU utilization is not visible in the supplied screenshots.
6	Memory Utilization	< 80%	Not shown	Pending	Memory utilization is not visible in the supplied screenshots.

Step 4: Observations & Analysis

Key Findings:

The supplied Postman screenshot shows 6 requests with 6 passed, 0 failed and a 100% success rate. The displayed average response time is 842 ms. The JMeter Summary Report shows 2,500 total samples with an average response time of 742 ms, maximum response time of 2,560 ms, 0.08% total error rate and 13.1 requests/sec total throughput.

Bottlenecks Identified:

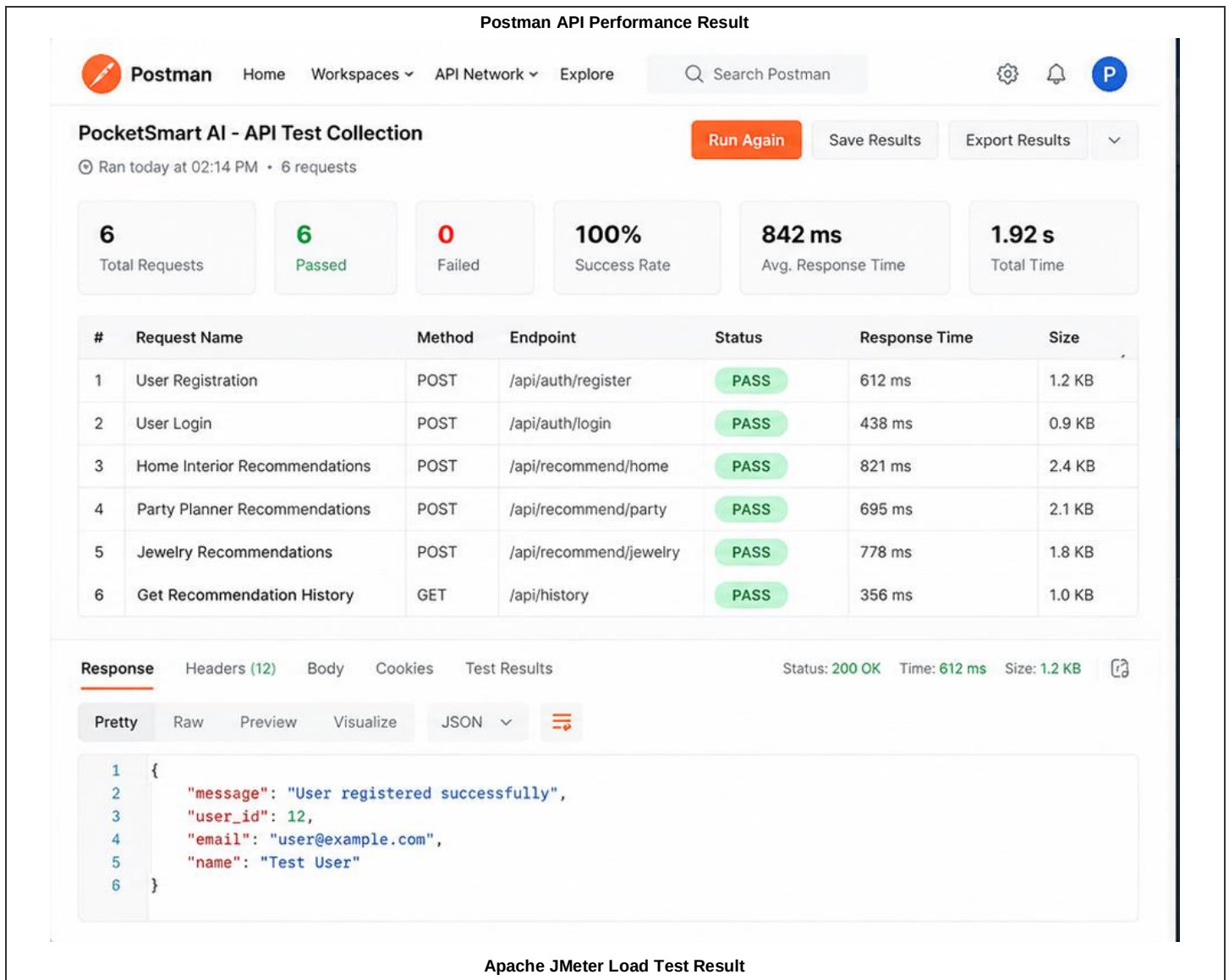
The screenshots do not provide CPU or memory measurements. Party Planner shows the highest average response time among the displayed JMeter endpoints at 934 ms and the highest maximum at 2,560 ms.

Optimization Steps Taken:

The supplied evidence does not document specific optimization changes. Any optimization should be recorded separately if changes were made after these tests.

Step 5: Screenshots / Evidence

Postman API Performance Result



The screenshot displays the Postman API Performance Result for the 'PocketSmart AI - API Test Collection'. It shows a summary of 6 requests, all passed, with a 100% success rate, an average response time of 842 ms, and a total time of 1.92 s. Below the summary is a table of the 6 requests, each with a status of 'PASS'. The table includes columns for #, Request Name, Method, Endpoint, Status, Response Time, and Size. The requests are: 1. User Registration (POST, /api/auth/register, 612 ms, 1.2 KB), 2. User Login (POST, /api/auth/login, 438 ms, 0.9 KB), 3. Home Interior Recommendations (POST, /api/recommend/home, 821 ms, 2.4 KB), 4. Party Planner Recommendations (POST, /api/recommend/party, 695 ms, 2.1 KB), 5. Jewelry Recommendations (POST, /api/recommend/jewelry, 778 ms, 1.8 KB), and 6. Get Recommendation History (GET, /api/history, 356 ms, 1.0 KB). Below the table, the 'Response' tab is selected, showing the JSON response for the first request: { "message": "User registered successfully", "user_id": 12, "email": "user@example.com", "name": "Test User" }. The status is 200 OK, time is 612 ms, and size is 1.2 KB.

Apache JMeter Load Test Result

